

Haitham Ismail

Machine Learning Engineer | Beirut, Lebanon | 0096171060782 | haithamismail.github.io | haithamismail.hi@gmail.com

SUMMARY

Highly skilled Computer Engineer with a strong foundation in software development and machine learning. Proven expertise in Python, TensorFlow, and data analysis, with a deep understanding of machine learning algorithms and neural networks. Demonstrated ability to manage complex projects and solve challenging problems. Committed to continuous learning and staying updated with the latest AI technologies and trends. Looking forward to applying my skills and knowledge to contribute to innovative AI solutions.

EDUCATION

Bachelor of Engineering in Computer Engineering, Al Maaref University, October 2018 - June 2022

- Graduated with high distinction, consistently listed on the Dean's Honor List
- GPA: 3.96/4.0

SKILLS

- Software: Python, TensorFlow, Torch, OpenCV, C, C++, MATLAB
- Soft Skills: Excellent communication, problem-solving, critical thinking, teamwork, and time management skills.

EXPERIENCE

Machine Learning and Data Engineer, boxMind.ai, Lebanon | November 2023 - April 2025

- Developed an end-to-end automated call center that utilizes generative AI and MS Azure services to answer the customers' questions in a human-like fashion
- Developed a video search system that indexes the videos using a metadata enrichment pipeline and enables semantic search on clips from videos for user queries (Local and Azure versions)
- Optimized and refactored existing GenAI solution codebase for better RAG results and ease of development
- Fine-tuned large language models using state-of-the-art techniques including QLoRA and Deep Speed for fast and efficient training
- Assisted in building a data pipeline to sync a storage directory and ingests different types of documents into a vector database
- Deployed different types of models such as LLMs, VLMs, stable diffusion, and embedding models, and integrated them into the system
- Technologies Used: PyTorch, HuggingFace Transformers, LlamaIndex, LangChain, LangGraph, Docker, Airflow, Flask, Milvus, MS Azure, GoLang, Linux

AI Engineer, Freelancing | May - July 2024

- Chose server specifications based on business requirements like generation speed, number of concurrent users
- Deployed Qwen2.5-VL to classify different documents, OCR, and structured information extraction from documents with over 95% accuracy
- Used OpenCV and pyzbar to extract barcodes with near perfect accuracy
- Integrated all functionalities into a FastAPI server

AI Business Analyst, Maids.cc | April 2025-Present

- Decreased the cost of chatbots by around 30% by implementing prompt caching
- Automated several workflows using LangGraph Agents like generating flowcharts for business specifications, verifying chatbot functionality according to business specification files
- Worked on automated alerting system using OpenSearch
- Ported chatbots to N8N with mongodb as database
- Worked on improving prompt for chatbots
- Proposed using tool in chatbots which improved the chatbots accuracy and maintainability

Automation Engineer Freelancing | June 2025

Automated workflows using N8N including:

- Generating email reminders for pending deals
- Fixing and syncing data between CRM and Supabase
- Syncing people to supabase and pipedrive using data from fireflies

Machine Learning Engineer, Freelancing, KATALYST, Remote Germany | June 2024

- Implemented feature to ingest and parse images, videos, pdfs, and text files to the system
- Extracted description from images and videos using state-of-the-art VLMs
- Implemented a RAG system with a multi-modal database and open clip embeddings
- Technologies used: openCV, LanceDB, CLIP, LangChain, nltk

AI Bootcamp Instructor, youbee.ai | January 2024 - Present

- Created comprehensive explainer notebooks designed to be self-sufficient and to provide practical examples
- Recorded standalone sessions to explain each lesson
- Explained the lessons in live sessions to make the bootcamp more engaging
- Technologies used: Jupyter Notebooks, Markdown, Python, Numpy, Tensorflow, Pandas, Sci-kit Learn, Matplotlib
- [HaithamIsmail/ai-notebooks](https://haithamismail.com/ai-notebooks)

Embedded Software Engineer, Eigentec, Remote Germany | June 2022 - September 2023

- Ported projects to Real-Time operating system, prioritizing functionalities into tasks and interrupts
- Designed and edited embedded software in C/C++ languages for NXP microcontrollers like i.MX RT1170-1160

- Ported Machine Learning pipelines to the microcontrollers using packages like CMSIS DSP and TensorFlow Lite micro
- Set up communication between the microcontrollers and other devices using SPI, UART, and Ethernet
- Set up Direct Memory Access for high speed data transfer between the peripherals
- Worked with several other boards including Analog Front-Ends and Freedom Boards
- Developed Python applications for user control of embedded software with data logging capabilities
- Documented projects for user-friendly customer interaction
- Technologies Used: Python, C, C++, NXP MCUXpresso, NXP FreeMASTER, NXP Microcontrollers, TF Lite micro

LANGUAGES

- Arabic: Native proficiency
- English: Full professional proficiency

CERTIFICATIONS

[Huawei Certified ICT Associate in Cloud Computing](#)

Certificate ID: 01010080280806277763629

[Deep Learning Specialization by DeepLearning.ai](#)

[TensorFlow: Advanced Techniques](#)

PUBLICATIONS

- [Internet-of-Things-Based Smart-Home Time-Priority-Cost \(TPC\)-Aware Energy Management System for Energy Cost Reduction](#)